WEEK 9 · LECTURE L1 · UNIT III

Algorithmic Decision Making in HR

Where algorithms now shape hiring, appraisal and work, why HR is harder than other domains, and how algorithms control work.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Map where algorithms make or shape HR decisions, the specific risk each carries, and what makes algorithmic decisions distinctive; explain algorithm aversion and appreciation.
2. Explain why AI in HR is harder than in other domains (Tambe, Cappelli and Yakubovich).
3. Analyze algorithmic control with Kellogg, Valentine and Christin’s six Rs, and learn from the Amazon case.
4. Frame AI risk with the NIST AI RMF and ISO/IEC 23894, and explain why fairness checks in HR rest on judgments, not facts.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 3 and CLO 5.

COMMIT BEFORE THEORY

Commit to an answer first

A firm trains a hiring model on ten years of its own hiring decisions, during which it hired mostly men. What should it expect?

A

A neutral model, because algorithms have no prejudice

B

A model that learns to prefer what past decisions preferred, including their bias

C

A model that corrects past bias automatically

D

A model too inaccurate to use

EXAM LENS

The answer is B.

WHERE ALGORITHMS ACT IN HR

Six arenas, and the stakes

Screening and matching

Ranking applicants and matching people to roles.

Appraisal

Scoring performance from activity data.

Promotion and pay

Recommending who moves up or earns more.

Attrition prediction

Scoring who is likely to leave.

Scheduling

Assigning shifts and tasks.

Algorithmic management

Directing, rating and disciplining platform workers.

EXAM LENS

The higher the stakes for a person, the higher the bar of evidence and accountability.

Four components, and what is distinctive

Four components

Input data about people; the model that transforms inputs; the output (prediction, recommendation, score or decision); and the action humans or systems take on it.

What is distinctive

Scale (thousands of cases at once) and consistency (the same logic every time). That removes arbitrary inconsistency, and repeats any error identically across every case.

TAMBE, CAPPELLI AND YAKUBOVICH (2019)

Why AI in HR is harder

1 · Complex phenomena

“Good performance” is hard to define and measure.

2 · Small data

Few hires or exits compared with consumer data.

3 · Accountability

Fairness and legal limits on decisions about people.

4 · Employee reactions

People may reject decisions made by algorithms.

EXAM LENS

Their responses: causal reasoning rather than correlation; randomization where a choice is genuinely uncertain; formal processes employees can understand and contribute to.

KELLOGG, VALENTINE AND CHRISTIN (2020)

Algorithms at work: six Rs of control

Mechanism	Two Rs	What it does
Direct	Restricting, recommending	Limits and steers what workers can do
Evaluate	Recording, rating	Tracks and scores every action
Discipline	Replacing, rewarding	Removes or rewards workers by score

EXAM LENS

Algorithmic control is more comprehensive, instantaneous, interactive and opaque than supervision. Workers respond with “algoactivism”: resisting or working around the algorithm.

THE AMAZON CASE (DASTIN, 2018)

How the past becomes the rule

Training data

A decade of resumes from a male dominated hiring history.

What it learned

Patterns linked to past hires, including words linked to women.

What it did

Downgraded resumes mentioning “women’s”.

Outcome

Engineers could not guarantee neutrality; the project was dropped.

EXAM LENS

Historical bias: perfect prediction of a biased past produces biased decisions; correcting it is a normative choice, not a technical one. A model trained on past decisions inherits their bias unless someone checks.

EVERY HR ALGORITHM CARRIES A SPECIFIC RISK

Name it

Decision	What the algorithm does	The risk
Screening	Ranks or filters applicants	Learns past bias; unknown validity
Matching	Suggests roles, shifts or tasks	Feedback loops that lock people in
Appraisal	Scores performance from activity data	Measures what is easy, not what matters
Promotion	Ranks readiness	Proxies for tenure, gender or location
Attrition prediction	Flags who may leave	False alarms that change how people are treated

EXAM LENS

Precision can be equal while error rates differ sharply: in an illustrative model, both groups have 25% precision, yet one group’s stayers are wrongly flagged at 47% against 18% (worked in Lecture 2).

THEORY IN DEPTH · DIETVORST, SIMMONS AND MASSEY (2015)

People abandon algorithms after one mistake

The core idea

Algorithm aversion: after seeing a model err, people prefer human judgment even when the model is more accurate.

How it works

People forgive human errors more readily than machine errors.

Limits and critique

Letting people slightly adjust a model’s output reduces aversion.

IN PRACTICE

Managers who see one bad flight risk flag may stop trusting all analytics, good and bad alike.

EXAM LENS

Pair with Logg et al. to show trust in algorithms can err in both directions.

THEORY IN DEPTH · LOGG, MINSON AND MOORE (2019)

People also over trust algorithms

The core idea

Algorithm appreciation: lay people often weight advice more when told it comes from an algorithm.

How it works

Appreciation is stronger for numeric estimates and weaker among experts.

Limits and critique

Aversion and appreciation coexist; which wins depends on the task and the person.

IN PRACTICE

A flight risk score shown as a precise number may be trusted more than it deserves; show its error rate beside it.

EXAM LENS

A strong answer reconciles aversion and appreciation rather than citing only one.

THE MEASUREMENT PROBLEM

Fairness checks assume we know the true outcome

Judgments, not facts

In HR, “good performance” and “qualified” are partly judgments; a manager rating is a judgment, an exit is a recorded fact.

Imperfect proxies

Measurements stand in for constructs; biased ratings standing in for performance build bias into the target.

Outcomes shaped by the decision

A rejected candidate never gets the chance to perform, so the counterfactual is never observed.

Course’s AI in HR Decisions teaching note.

NIST (2023); ISO/IEC 23894 (2023)

Framing AI risk

Three kinds of harm (NIST)

To people: individuals, groups, society. To an organization: operations, security, money, reputation. To an ecosystem: interconnected systems and supply chains.

Life cycle (NIST)

Plan and design; collect and process data; build and use the model; verify and validate; deploy; operate and monitor. Testing, evaluation, verification and validation run through every stage.

ISO/IEC 23894

Guidance used with ISO 31000: principles, framework and process. Rights risks (equality, privacy, dignity, autonomy) treated as hard constraints; treatment options: avoid (no AI for terminations), reduce, share or accept within risk appetite.

ISO risk categories and treatment options as given in the course’s teaching note; the filed ISO preview does not contain those clauses.

APPLY

Algorithmic shift scheduling in a warehouse

Teaching case (Course framing)

A Bhiwandi warehouse uses an algorithm that assigns shifts, sets picking targets, ranks workers daily and automatically issues warnings to the bottom 5%. Workers cannot see how targets are set. Some have started skipping safety checks to raise their pick rates.

Work it through

1. Which of the six Rs does the system use?
2. Why is this control more intense than a supervisor's?
3. Which of Tambe and colleagues' four challenges apply?
4. How would you explain the skipped safety checks?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Algorithms are neutral because they have no feelings	They learn from past decisions and data, including bias
AI works in HR as well as in marketing	HR data are small, outcomes hard to measure, decisions legally constrained
Algorithmic control is just faster supervision	It is more comprehensive, instantaneous, interactive and opaque
Consistency means fairness	Consistency repeats any error identically, in every case

COMPARE AND EVALUATE

Five lenses on algorithmic HR

Lens	Question it answers	Watch out for
Tambe et al. (2019)	Why is AI in HR hard?	Remedies are demanding
Kellogg et al. (2020)	How do algorithms control work?	Built mostly on platform cases *
Amazon (Dastin, 2018)	How does history become a rule?	A press report, one firm
NIST AI RMF (2023)	How should risk be organized?	Voluntary
ISO/IEC 23894 (2023)	How does AI risk fit enterprise risk?	Process guidance, not thresholds

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

A Bengaluru logistics platform uses algorithms to allocate orders, rate riders and deactivate low scorers. Using Kellogg, Valentine and Christin and Tambe, Cappelli and Yakubovich, evaluate this system and propose two safeguards. **20 marks**

A strong answer: maps the system to direction, evaluation and discipline through the six Rs; explains why algorithmic control differs from supervision; applies two of the four challenges; proposes safeguards such as transparency, human review and redress, with a limit.

“A model trained on a firm’s past decisions will reproduce its past.” Discuss with the Amazon case and the measurement problem in HR. **20 marks**

A strong answer: explains historical bias; uses Amazon; explains judgments as targets and unobserved counterfactuals; proposes checks.

Name NIST’s four functions and explain which one an audit belongs to. **10 marks**

A strong answer: Govern, Map, Measure, Manage; an audit is Measure, and needs Govern and Manage to matter.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Higher stakes, higher bar.**
   Evidence and accountability rise with the stakes for a person.

2. **HR is harder for AI.**
   Complex outcomes, small data, accountability, reactions.

3. **Algorithmic control is total and opaque.**
   Direct, evaluate, discipline.

4. **The past becomes the rule.**
   Unless someone checks the data and the target.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Tambe, P., Cappelli, P., & Yakubovich, V. (2019). AI in HRM. *CMR, 61*(4).
- Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work. *AOM Annals, 14*(1).
- Leicht-Deobald, U., et al. (2019). *JBE, 160*(2).

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion. *Journal of Experimental Psychology: General, 144*(1), 114 to 126. https://doi.org/10.1037/xge0000033
- Logg, J. M., Minson, J. A., & Moore, D. A. (2019). Algorithm appreciation. *Organizational Behavior and Human Decision Processes, 151*, 90 to 103. https://doi.org/10.1016/j.obhdp.2018.12.005
- Dastin, J. (2018, October 10). Amazon scraps secret AI recruiting tool that showed bias against women. Reuters.
- ISO/IEC. (2023). *ISO/IEC 23894:2023 Information technology: Artificial intelligence: Guidance on risk management*.
- Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals, 14*(1), 366 to 410. https://doi.org/10.5465/annals.2018.0174
- National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)*.
- Tambe, P., Cappelli, P., & Yakubovich, V. (2019). Artificial intelligence in human resources management: Challenges and a path forward. *California Management Review, 61*(4), 15 to 42. https://doi.org/10.1177/0008125619867910